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Addressing BCI inefficiency in c-VEP-based BCIs: A comprehensive study of neurophysiological predictors, binary stimulus sequences, and user comfort

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Keywords: brain-computer interface (BCI), code-modulated visual evoked potential (c-VEP), electroencephalography (EEG), neurophysiological predictors, reconvolution, BCI inefficiency, binary stimulus sequences

Abstract

Objective. This study investigated the presence of brain-computer interface (BCI) inefficiency in BCIs using the code-modulated visual evoked potential (c-VEP). It further explored neurophysiological predictors of performance variability and evaluated a wide range of binary stimulus sequences in terms of classification accuracy and user comfort, aiming to identify strategies to mitigate c-VEP BCI inefficiency. *Approach.* In a comprehensive empirical analysis, ten different binary stimulus sequences were offline evaluated. These sequences included five code families (m-sequence, de Bruijn sequence, Golay sequence, Gold code, and a Gold code set), each in original and modulated form. To identify predictors of performance variability, resting-state alpha activity, heart rate and heart rate variability, sustained attention, and flash-VEP characteristics were studied. *Main Results.* Results confirmed substantial inter-individual variability in c-VEP BCI efficiency. While all participants reached a near-perfect classification accuracy, their obtained speed varied substantially. Four flash-VEP features were found to significantly correlate with the observed performance varibility: the N2 latency, the P2 latency and amplitude, and the N3 amplitude. Among the tested stimulus conditions, the m-sequence emerged as the best-performing universal stimulus. However, tailoring stimulus selection to individuals led to significant improvements in performance. Cross-decoding was successful between modulated stimulus conditions, but showed challenges when generalizing across other stimulus conditions. Lastly, while overall comfort ratings were comparable across conditions, stimulus modulation was associated with a significant decrease in user comfort. *Significance.* This study challenges the assumption of universal efficiency in c-VEP BCIs. The findings highlight the importance of accounting for individual neurophysiological differences and underscore the need for personalized stimulus protocols and decoding strategies to enhance both performance and user comfort.

1. Introduction

A brain-computer interface (BCI) enables direct communication between the brain and external devices by recording and interpreting neural activity [1]. Various neural signals can be used to drive a BCI, including oscillatory responses such as those induced by motor-imagery (MI), and evoked responses like the event-related potential (ERP), steady-state visual evoked potential (SSVEP) and code-modulated visual evoked potential (c-VEP) [2].

MI-based BCIs use changes in spontaneous brain oscillations when a user imagines or attempts a movement [3]. This is contrasted to the evoked protocols, which rely on external stimulation. ERP protocols typically operate at a relative slow pace, evoking not only early sensory responses, but also more downstream cognitive processes [4]. Instead, SSVEP protocols employ rapid periodic frequency-tagging and rely on early sensory responses [5]. Similarly, c-VEP relies on early visual responses, but those following rapid non-periodic noise-tagging [6].

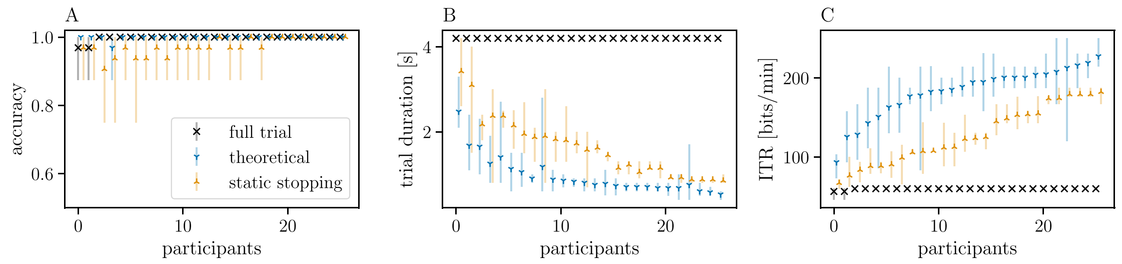

Figure 1. c-VEP BCI inefficiency. Displayed are the average 4-fold cross-validated accuracy (A), trial duration (B), and information transfer rate (ITR, C) for individual participants using a c-VEP BCI. Bars represent the range across folds. (Black) Metrics for the full 4.2 s trials, where no BCI inefficiency is observed. (Blue) Metrics at the shortest trial durations where maximum performance is reached, highlighting substantial variability in ITR across participants. (Yellow) Metrics using a static stopping procedure that minimizes trial duration to reach maximum accuracy based on training data. Participants are ordered by theoretical ITR (blue).

The rapid and non-periodic nature of c-VEP stimulation enables a higher information bandwidth than traditional methods [6, 7]. Consequently, c-VEP is gaining attention as a promising alternative to the well-established MI, ERP, and SSVEP-based BCIs. Recent studies have demonstrated c-VEP's excellent performance in speller BCIs, achieving remarkable information transfer rates (ITRs) of over 220 bits/min [7, 8].

1.1. BCI inefficiency

Within the BCI research community, it is widely acknowledged that BCIs do not work well for all users. It was reported, and is frequently reiterated, that BCI control fails for approximately 15–30% of users [9–12]. A BCI is typically considered uncontrollable when its classification accuracy falls below 70% [9]. Originally, this phenomenon was termed BCI illiteracy, emphasizing the user's inability to achieve control. More recently, the term BCI inefficiency was introduced to shift the focus toward the system's limitations rather than attributing failure to the user [13].

A substantial body of research has sought to identify the underlying factors contributing to BCI inefficiency. Identified factors range from demographic variables such as age and gender to cognitive and emotional states, as well as personality traits and cognitive profiles [14]. In addition, numerous studies have investigated neurophysiological predictors, with the μ-rhythm power during an eyes-open resting state emerging as one of the most consistently reported predictors [11, 15].

Recently, the notion of BCI inefficiency has been challenged. A study reported that all 86 individuals were able to successfully use a c-VEP BCI [16]. Similarly, BCIs using other evoked responses report a very low if not non-existing BCI inefficiency rate [10, 17–19]. This may suggest that the originally defined BCI inefficiency may be primarily associated with induced responses such as the MI paradigm. That is, the commonly cited statistic may more accurately be stated as: MI-based BCI control fails for 15–30% of users.

However, when interpreting BCI inefficiency as a continuous scale, with users reaching different levels of BCI control, rather than classifying into successful and unsuccessful control, then one can also observe BCI inefficiency among evoked response BCIs, such as those using c-VEP. Specifically, all c-VEP BCI users can reach a near perfect classification accuracy, here between 0.969–1.000, with sufficient data, here 4.2 s (figure 1 full trial). Crucially, when minimizing trial durations to reach the same maximum accuracy, one can see that some users need only 500 ms, while others require 2.5 s trials (figure 1 theoretical). Consequently, the ITR varies from 93.5 bits/min to 227.3 bits/min across users. That is, while all users may achieve high accuracy, their efficiency may still vary substantially.

Acknowledging that c-VEP BCIs also exhibit some degree of inefficiency—or various degrees of efficiency—the original question remains highly relevant: What causes these inter-individual differences? This study explores the much reproduced neurophysiological predictor from Blankertz *et al* [11], here applied on visual alpha. Additionally, this study examines heart rate variability (HRV), which was shown to correlate with the performance of a P300 BCI [20]. Finally, this study investigates various flash-VEP characteristics—the response to a single flash, which forms the basis of the c-VEP, which is the response to a random sequence of flashes—which has been shown to vary considerably between individuals [21–24].

1.2. Stimulus sequences for c-VEP BCI

If users respond differently to c-VEP stimuli, given their underlying VEP, instead of seeking a universal stimulus protocol that maximizes average performance across users, a more effective approach may be to tailor stimulation to each individual. This personalized strategy could enhance BCI performance and potentially reduce the observed variability in BCI efficiency.

Most c-VEP BCIs use binary sequences from telecommunications due to their advantageous properties, such as near-zero auto- and cross-correlation. These include m-sequences [25–28], Gold codes [29–32], Barker codes [33], Golay sequences [31, 34, 35], almost-perfect autocorrelation (APA) sequences [31, 35, 36], and de Bruijn sequences [31, 35].

A crucial limitation of these sequences is often overlooked: They are optimized for digital signal transmission, whereas BCI performance depends on optimal correlation properties within the response domain. The assumption that desirable properties in the stimulus domain directly translate to the response domain is not always valid [35, 37].

These binary sequences are typically presented at full contrast, alternating between black and white, which may maximize the signal-to-noise ratio (SNR), but is often perceived as uncomfortable by users. To improve user experience, studies explored non-binary sequences that alternate between different levels of gray, such as non-binary m-sequences [26, 38].

More recently, researchers have begun optimizing codes to achieve desirable properties. Examples are periodic, quasi-periodic, and chaotic patterns [39, 40], superposition optimized (SOP) sequences [41] and discrete interval binary sequences (DIBS) [42]. Additionally, studies have sampled stimuli directly from random distributions [7, 8, 43–46], which allows for further optimization to improve BCI performance [44, 46].

Unfortunately, direct comparisons between different codes remain scarce and existing studies report mixed results. Isaksen *et al* found no differences between m-sequences, Gold codes, and Barker codes [33], in line with Shirzhiyan *et al* who found no differences between m-sequences and chaotic codes [39]. In contrast, Wei *et al* found Golay sequences outperforming m-sequences [34], Nagel *et al* found m-sequences outperforming random sequences [43], and Lai *et al* found m-sequences outperforming DIBS [25]. Finally, Behboodi *et al* showed time-factor optimum and 6-factor optimum codes outperforming m-sequences [47], and Castillos *et al* found burst codes outperforming m-sequences [48].

Only two studies conducted broader analyses, though both were based on simulated data. Torres and Daly found that APA, de Bruijn, and Golay sequences generally outperformed m-sequences, Gold codes and Kasami sequences [31]. Similarly, Thielen reported that Golay sequences performed best, followed by the APA sequence, m-sequences, Gold codes, and de Bruijn sequences [35].

With the vast number of available sequences for c-VEP BCIs, there is significant potential for optimization. Isaksen *et al* found no difference in average performance between m-sequences, Gold codes, and Barker codes. However, when selecting the optimal sequence for each individual, average accuracy improved from 87.2% to 95.7% [33].

1.3. Aims of the current study

This study has three primary objectives. First, this study shows that BCI inefficiency exists in c-VEP BCI. Specifically, this study demonstrates a large variance in the BCI efficiency obtained by participants, when treating BCI inefficiency beyond mere classification accuracy, but instead considering also classification speed and ITR.

Second, this study aims to identify factors that may contribute to this observed variability across users. Understanding these factors may offer insights into the underlying mechanisms driving individual differences in c-VEP BCI performance and may inspire strategies to mitigate them.

Third, acknowledging this inter-individual variability, the study aims to compare and optimize stimulus protocols by tailoring them to individual users. Given the crucial role of stimulus protocols in both BCI performance and user comfort, this research evaluates the effectiveness of a large set of commonly used binary stimulus sequences for c-VEP BCIs using an empirical dataset.

2. Methods

2.1. Participants

Twenty-six participants, with ages ranging from 18 to 31 yr (mean age 23 yr), participated in the experiment, comprising 17 females and 9 males. Exclusion criteria were a history of epilepsy or claustrophobia. All participants had either normal or corrected to normal vision, reported no abnormalities in the central nervous system, and were right-handed. Prior to the experiment, all participants provided written informed consent and received compensation for their contribution. The experimental procedures and methods were approved by and conducted in accordance with the guidelines established by the local ethical committee of the Faculty of Social Sciences at Radboud University, and the Declaration of Helsinki.

2.2. Materials

Electroencephalography (EEG) data were collected using 64 sintered Ag/AgCl active electrodes. The electrodes were positioned according to the 10-10 international system and amplified by a Biosemi ActiveTwo at a sampling rate of 2048 Hz.

Electrocardiography (ECG) data were collected using a Polar H10 heart rate sensor at a sampling rate of 130 Hz. The sensor was secured just below the participant's chest with a chest strap. The ECG data of one participant (sub-09) were missing due to an issue with the device.

A 1920×1080 px BenQ XL2420Z LED monitor was used for visual stimulation, placed 60 cm in front of participants. The stimuli were delivered at a presentation rate of 60 Hz, achieved through upsampling to match the monitor's 120 Hz frame rate, mitigating the adverse effects of monitor raster latency [49].

The monitor displayed either a single stimulus or a speller grid (figure 2). The single stimulus involved a 3×3 visual degree box with a fixation cross, centered

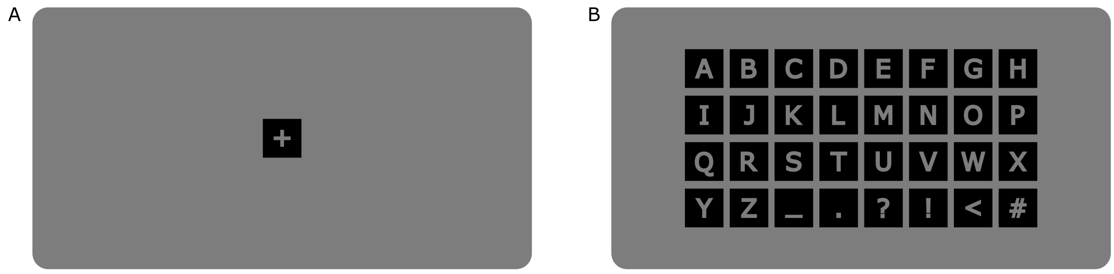

Figure 2. The stimulus screens. (A, B) Displayed are the screens for the single stimulus (A) and speller grid (B). Both were displayed centered on a mean-luminance gray background. Individual boxes were 3×3 visual degree. Boxes in the speller grid were separated with 0.5 visual degrees.

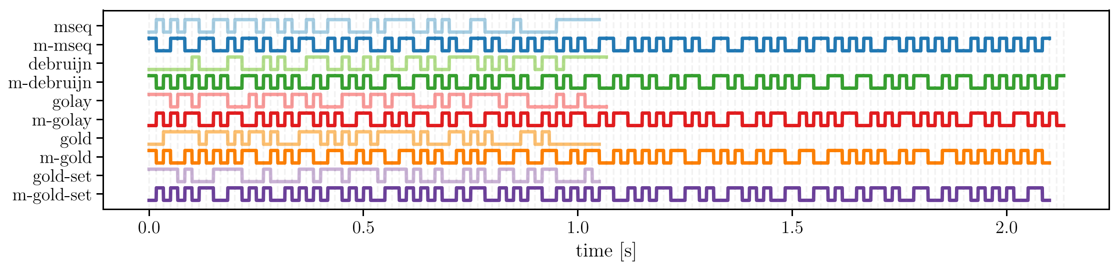

Figure 3. The stimulus conditions. The figure displays one cycle for five code families: a shifted m-sequence (mseq, blue), a shifted de Bruijn sequence (debruijn, green), a shifted Golay sequence (golay, red), a shifted Gold code (gold, orange), and a set of Gold codes (gold-set, purple). Each sequence is shown either in its original form (light color) or as modulated (prefixed with m-, dark color). The vertical gray lines indicate individual bits at a presentation rate of 60 Hz. Note that the modulated sequences are twice as long. However, during the experiment, multiple repetitions of these cycles were presented to match a common trial duration across conditions.

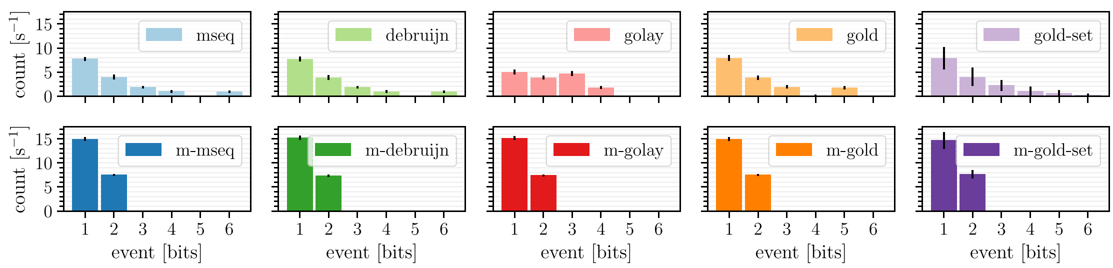

Figure 4. The event distributions across stimulus conditions. Displayed are the event distributions for the five code families: m-sequence (mseq, blue), de Bruijn sequence (debruijn, green), Golay sequence (golay, red), a Gold code (gold, orange), and a set of Gold codes (gold-set, purple). They were presented either in its original form (top row, light color) or as modulated (bottom row, dark color). Events are defined as consecutive runs of ones. Distributions are normalized to counts per second to account for varying cycle lengths.

on a mean-luminance gray background. The speller grid consisted of a 4×8 matrix, also centered on a mean-luminance gray background. The matrix contained 32 unique symbols, each within a 3×3 visual degree box, with a box-to-box spacing of 0.5 visual degrees.

2.3. Stimulus conditions

As part of the experiment, participants were presented with the 4×8 speller grid. Each cell in the grid was luminance-modulated at full contrast with pseudo-random bit-sequences from ten different stimulus conditions (figure 3). These conditions contained five distinct code families in their original and modulated form, such that their distribution of events, that is, the distribution of flash durations within a stimulus varied substantially (figure 4).

2.3.1. Shifted m-sequence

One code family was the widely used circularly shifted m-sequence [50]. An m-sequence was generated using a shift register of length 6 with feedback tap positions at 6 and 1. This configuration produced a code sequence of length $2^6 - 1 = 63$ bits, equivalent to

1.05 s at 60 Hz. This m-sequence was circularly shifted with a lag of 2 bits to create the other sequences.

The m-sequence was nearly balanced, containing one more one than zeros. The set contained five distinct flash durations: 16.67 ms (1 bit), 33.33 ms (2 bits), 50 ms (3 bits), 66.67 ms (4 bits), and 100 ms (6 bits).

This condition will be referred to as 'mseq'.

2.3.2. Shifted de Bruijn sequence

Another code family was a circularly shifted de Bruijn sequence [51]. The de Bruijn sequence was generated with an order of 6 and an alphabet size of 2, resulting in a sequence that contained all possible binary subsequences of length 6 exactly once. Consequently, the sequence had a length of $2^6 = 64$ bits, about 1.07 s at 60 Hz. The sequence was circularly shifted by 2 bits increments to generate the full set.

The de Bruijn sequence had an equal number of ones and zeros. The set included five distinct flash durations: 16.67 ms (1 bit), 33.33 ms (2 bits), 50 ms (3 bits), 66.67 ms (4 bits), and 100 ms (6 bits).

This condition will be referred to as 'debruijn'.

2.3.3. Shifted golay sequence

Another code family was a circularly shifted Golay sequence [52]. Golay sequences are generated in pairs, known as complementary sequences, ensuring that their out-of-phase aperiodic autocorrelation sums to zero. A Golay sequence of order 6 creates a pair of codes with a length of $2^6 = 64$ bits, which is about 1.07 s at 60 Hz. The Golay sequence used in this study was taken from the study of Wei *et al* [53], and was circularly shifted by 2 bits to produce the remaining codes.

The Golay sequence contained 36 ones and 28 zeros, and four distinct flash durations ranging from 16.67 ms (1 bit) to 66.67 ms (4 bits).

This condition will be referred to as 'golay'.

2.3.4. Set of Gold codes

Another code family was a set of Gold codes [54]. Gold codes are generated by XOR-ing a preferred pair of m-sequences across all possible lags. Both m-sequences were created with a 6-bit shift register, with feedback tap positions at 6 and 1 for the one sequence, and 6, 5, 2, and 1 for the other. This procedure resulted in a set of $2^6 - 1 = 63$ Gold codes, each consisting of $2^6 - 1 = 63$ bits, equivalent to 1.05 s at 60 Hz.

Among the set of 63 sequences, only 39 had a maximum run-length of 6 and were nearly balanced, with one more ones than zeros. From these 39, the initial 32 were selected. This subset of Gold codes contained all six flash durations ranging from 16.67 ms (1 bit) to 100 ms (6 bits).

This condition will be referred to as 'gold-set'.

2.3.5. Shifted gold code

The last code family was a circularly shifted Gold code. For this, the first code from the gold-set was used. This code was cyclically shifted by 2 bits to create the remaining codes.

The shifted Gold code was almost balanced with one more one than zeros. The set contained all five flash durations from 16.67 ms (1 bit) up to 83.33 ms (5 bits).

This condition will be referred to as 'gold'.

2.3.6. Modulated sequences

Each of the five aforementioned code families was presented in both its original form and a modulated version. The modulation limited the number of flash durations, shifting the frequency content to a higher band [55].

Modulation was applied using an XOR operation with a double-frequency bit-clock [55]. This process replaced every 0 in a sequence with 10 and every 1 with 01. Consequently, sequences doubled in their length, with the modulated m-sequence and Gold codes extending to 2.1 s, and the modulated de Bruijn and Golay sequences approximately 2.13 s.

This modulation ensured that each code contained an equal number of ones and zeros while restricting the run-length distribution to a maximum of two. That is, modulated codes exclusively contained flash durations of 16.67 ms (1 bit) and 33.33 ms (2 bits).

Modulated conditions will be appended with 'm-', creating five more stimulus conditions: m-mseq, m-debruijn, m-golay, m-gold, m-gold-set.

2.4. Experiment

The experiment consisted of a sequence of tasks performed in the following chronological order: a resting-state task, a VEP task, a c-VEP task, a second resting-state task, an attention task, and finally, a comfort task (figure 5). All tasks were completed within a single EEG session, with the entire procedure, including also setup, briefing, and debriefing, taking no more than two hours.

2.4.1. Resting-state task

Participants completed two resting-state tasks, one before and one after the main experiment. Data collected during these tasks can be used to estimate various neurophysiological variables that may predict BCI performance.

During the resting-state task, the single stimulus was displayed and participants were instructed to fixate on the central cross. Each resting-state task consisted of two runs, one with eyes open and one with eyes closed, each run lasting 2 min.

2.4.2. VEP task

Participants completed one VEP task designed to reliably estimate the flash-VEP, the brain's response to

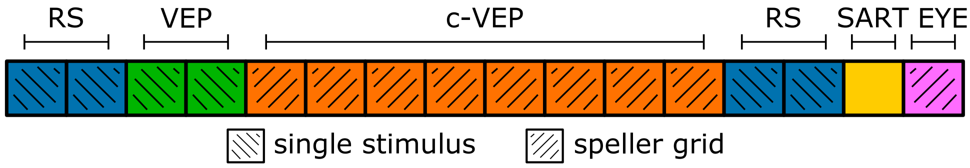

Figure 5. The experiment timeline. Illustrated is the experimental design used in this study. Participants completed five tasks in a single EEG session: a resting-state (RS) task, a visual evoked potential (VEP) task, a code-modulated VEP (c-VEP) task, a sustained attention to response task (SART), and a user comfort questionnaire on eye strain (EYE). To allow for breaks, tasks were divided into multiple runs, represented by individual boxes. Each task presented either the single stimulus screen or the speller grid (figure 2).

a single flash. Since the flash-VEP underlies the c-VEP, individual differences in flash-VEP characteristics may help explain variability in c-VEP performance.

During the task, the single stimulus was displayed, and participants were instructed to fixate on the central cross. The task consisted of two runs, each comprising 45 trials. Trials had an average duration of 5.2 s, with a 2 s inter-trial interval. The order of trials was randomized across participants.

Each trial presented 13 flashes, covering six different flash durations ranging from 16.67 ms (1 bit) to 100 ms (6 bits) in 16.67 ms increments. Each duration appeared twice, with an additional 16.67 ms flash completing the sequence. The order of flash durations was randomized using a balanced Latin square. Consecutive flashes were separated by a uniformly sampled inter-stimulus interval ranging between 250–500 ms, minimizing response overlap and effectively jittering potential cross-coupling between flashes.

In summary, participants underwent a total of 90 trials. Among these, the flash duration of 16.67 ms was presented 270 times, while the other five flash durations occurred 180 times, resulting in a total of 1170 flashes.

2.4.3. c-VEP task

Participants completed a c-VEP task involving various stimulus conditions, allowing for the estimation of c-VEP BCI performance across different stimulus sequences.

During this task, a 4×8 speller grid was displayed, and participants were instructed to fixate on a target symbol. For each of the ten stimulus conditions, one trial was recorded for every symbol serving as a target, resulting in a total of 320 trials. These trials were distributed across eight runs, each containing 40 trials. Participants could take self-paced breaks between runs. The order of trials and conditions was randomized across participants using a balanced Latin square.

Each trial began with a cue, highlighting the target symbol in green for 800 ms. This was followed by all symbols flashing according to the stimulus sequence of one condition for either 4.2 s (for m-sequences and Gold codes) or 4.27 s (for de Bruijn and Golay sequences). Consecutive trials proceeded without an additional inter-trial interval, other than the cue period.

2.4.4. Attention task

Participants completed the sustained attention to response task (SART) to assess sustained attention [56], which was explored as a potential predictor of c-VEP BCI performance.

During the task, digits from 1 to 9 were rapidly presented in white at the center of a black screen. Participants were instructed to press the spacebar as quickly as possible upon seeing a digit, except when the digit was a 3, in which case they were to withhold their response. Each digit appeared for 250 ms, followed by a mask displayed for 900 ms. Participants received feedback whenever they made an incorrect response.

The task began with a practice round, in which each digit was shown twice. In the main task, each digit appeared 25 times in a randomized order, ensuring that no digit was repeated consecutively.

2.4.5. Comfort task

Participants completed a brief task to evaluate the perceived eyestrain for each stimulus condition, providing a measure of the comfort level of different stimulus sequences.

During the task, the speller grid was displayed, allowing participants to freely view the stimulus interface. They were instructed to rate their eyestrain on a 7-point scale, where 1 indicated no eyestrain and 7 indicated significant eyestrain. Each stimulus condition presented one trial lasting 4.2 s. Participants could replay a trial or submit their rating using the keyboard, which then triggered the next condition. The order of conditions was counterbalanced across participants, and previous responses could not be modified.

2.5. Analysis

2.5.1. c-VEP decoding

In this study, all decoding was performed using the established and data-efficient reconvolution method [55]. Reconvolution decomposes responses to sequences of events (i.e., c-VEP) into responses to individual events (i.e., VEP). Beyond modeling the temporal response, the approach also learns a spatial filter by integrating reconvolution into a canonical correlation analysis (CCA) [32, 57].

Reconvolution CCA operates at the trial level. Each stimulus sequence i, where $i = 1, \ldots, 32$, is represented by an event time-series $\mathbf{E}_i \in \mathbb{R}^{E\times T}$, where E is the number of distinct event types considered in CCA, and T is the number of temporal features. In this study, $E = 3$ event types were modeled: the flash onset (i.e., '01' bits), the flash offset (i.e., '10' bits), and the initial stimulation onset at the beginning of the trial.

Each event time-series $\mathbf{E}_i$ is transformed into a structure matrix $\mathbf{M}_i \in \mathbb{R}^{M\times T}$, where M denotes the number of event-related time points. Assuming equal-length responses for each event, $M = E \times L$ with $L = 36$, corresponding to 300 ms at 120 Hz, sufficient to capture the flash-VEP. The structure matrix uses a Toeplitz representation to model the onset, duration, and overlap of responses to each event in sequence i.

During calibration, a CCA is fit that learns both a spatial filter $\mathbf{w} \in \mathbb{R}^C$ and a temporal filter $\mathbf{r} \in \mathbb{R}^M$ using supervised data. All K trials in the training set are concatenated along the temporal axis, which yields $\mathbf{S} = [\mathbf{X}_1, \ldots, \mathbf{X}_K]$. Similarly, the corresponding structure matrices are concatenated, which gives $\mathbf{D} = [\mathbf{M}_{y_1}, \ldots, \mathbf{M}_{y_K}]$.

The CCA optimization seeks to maximize the correlation ρ between the spatially and temporally filtered signals:

$$\underset{\mathbf{w},\mathbf{r}}{\arg\max} \quad \rho(\mathbf{w}^\top \mathbf{S}, \mathbf{r}^\top \mathbf{D}) \tag{1}$$

During inference, reconvolution CCA identifies the attended target symbol $\hat{y}$ by selecting the stimulus sequence i whose template response maximizes the correlation with the EEG signal:

$$\hat{y} = \underset{i}{\arg\max} \quad \rho(\mathbf{w}^\top \mathbf{X}, \mathbf{r}^\top \mathbf{M}_i) \tag{2}$$

The implementation of reconvolution CCA is publicly available in the PyntBCI Python library [58].

2.5.2. c-VEP BCI inefficiency

To assess the presence of BCI inefficiency in c-VEP BCI, a standard decoding pipeline was applied to the EEG data from the c-VEP task. This analysis focused on the commonly used circularly shifted m-sequence, consisting of 32 trials of 4.2 s each.

The EEG data was band-pass filtered between 6 and 30 Hz, a range optimized for c-VEP [59, 60]. Next, the EEG was segmented into trials, starting 500 ms before stimulation onset and ending 500 ms after stimulation offset. These trials were then downsampled to 120 Hz and cropped to retain only the stimulation phase, removing the initial and final 500 ms. This preprocessing ensured accurate marker alignment and minimized spectral filtering artifacts.

Performance was evaluated using a chronological 4-fold cross-validation. That is, the order of trials was not shuffled to maintain the temporal order of the data. The classification model used the template-matching approach with the reconvolution CCA method, as explained before.

Three decoding pipelines were implemented. First, classification was performed using the entire 4.2 s trial duration. Second, a theoretical upper bound on the performance was estimated by generating a decoding curve on the testing data by incrementally increasing the trial duration in 100 ms steps until the full 4.2 s duration. The earliest time point at which maximum accuracy was achieved was identified. Third, a static stopping procedure was applied, which similarly estimated a decoding curve and determined the earliest time point with maximum accuracy, but did so on the training data, practically simulating an online BCI system.

Each pipeline produced an estimated classification accuracy and trial duration, averaged over the cross-validation folds. The information transfer rate (ITR) was then computed using the formula:

$$b = \left(\log_2(n) + p\log(p) + (1 - p)\log_2\frac{1 - p}{n - 1}\right)\frac{60}{t}$$

where b is the ITR in bits per minute, $n = 32$ is the number of classes, p is the classification accuracy, and t is the selection time, incorporating the trial duration plus the 800 ms inter-trial time used for the cue.

2.5.3. c-VEP BCI performance prediction

This study examined multiple factors that may influence c-VEP performance, including visual alpha, HR and HRV, sustained attention, and VEP characteristics.

Alpha This study investigated the neurophysiological predictor proposed by Blankertz *et al* [11], originally applied to the sensorimotor mu rhythm. Here, it is adapted to the visual alpha rhythm, as increased power in the 8–12 Hz range has been linked to reduced information processing [61] and inhibitory mechanisms in task-irrelevant regions [62], suggesting decreased alertness to visual stimuli.

The alpha predictor was estimated from 2 min of EEG data recorded during the pre-session eyes-open resting-state task. The data were band-pass filtered between 4 and 40 Hz and spatially filtered using a small Laplacian filter. Power spectral density was then computed at channel Oz, where the following equation was fitted using least squares optimization:

$$g(f;\, \lambda, \boldsymbol{\mu}, \boldsymbol{\sigma}, \mathbf{k}) = g_1(f;\, \lambda, \mathbf{k}) + g_2(f;\, \boldsymbol{\mu}, \boldsymbol{\sigma}, \mathbf{k})$$

$$g_1(f;\, \lambda, \mathbf{k}) = k_1 + \frac{k_2}{f^\lambda}$$

$$g_2(f;\, \boldsymbol{\mu}, \boldsymbol{\sigma}, \mathbf{k}) = k_3\Phi(f;\, \mu_1, \sigma_1) + k_4\Phi(f;\, \mu_2, \sigma_2)$$

where f is the frequency, λ is the $1/f$ exponent, and k_1 and k_2 represent the noise offset and amplitude, respectively. The parameters μ_1 and σ_1 define the mean and standard deviation of the Gaussian Φ modeling the alpha peak with amplitude k_3, while μ_2 and σ_2 represent the beta peak with amplitude k_4.

Like the mu predictor, the alpha predictor is then computed as $\max_f g_2(f;\, \boldsymbol{\mu}, \boldsymbol{\sigma}, \mathbf{k})$, which takes into account the noise floor defined in g_1 when estimating the alpha and beta peaks in g_2, which was shown to improve the correlation of the mu predictor with BCI performance [11].

HR and HRV This study examined heart rate (HR) and heart rate variability (HRV) as a physiological predictor of c-VEP BCI performance, which have previously been linked to P300 BCI performance [20].

The ECG data recorded during the pre-session eyes-open resting-state task were band-pass filtered between 5 and 15 Hz. Subsequently, the Pan-Tompkins algorithm was used to detect QRS complexes [63]. The HR was then calculated as the average interval between consecutive R waves, and the HRV was quantified using the root mean square of successive differences (RMSSD) between intervals of consecutive R waves [64].

SART This study examined sustained attention as a psychological predictor of c-VEP BCI performance. In a previous study, attention span was found to correlate with P300 BCI performance [65]. Here, the performance of a motor response to a frequent stimulus and a withheld motor response to a rare stimulus was used as a measure of sustained attention. From the SART task, both an individual's accuracy as well as their response times (RTs) were extracted as potential predictors.

VEP This study assessed flash-VEP characteristics as a neurophysiological predictor of c-VEP performance. In previous studies, inter-individual differences were shown with respect to the flash-VEP amplitude and latency [21, 24], anatomy [23], and across age [22, 66] and sex [22, 24].

For this analysis, the EEG data from the VEP task was used. The EEG data was band-pass filtered between 6 and 30 Hz. The data was then sliced into epochs starting 200 ms before flash onset until 500 ms after. Epochs were baseline corrected using the 200 ms pre-stimulus interval and downsampled to 120 Hz.

After preprocessing, the epochs were averaged to obtain a flash-VEP. From this response, the latency and amplitude of the negative peak between 50 and 100 ms, the positive peak between 75 and 125 ms, and the negative peak between 100 and 200 ms were estimated. These ranges allowed for the most dominant N2, P2, and N3 peaks in the flash-VEP [66].

2.5.4. c-VEP BCI stimulus optimization

This study systematically compared a diverse set of binary stimulus sequences, containing ten stimulus conditions derived from five code families, each presented in both original and modulated forms. For this analysis, the EEG data of the c-VEP task was used. Performance was evaluated using two key analysis methods: the theoretical upper-bound estimation via decoding curves estimated from testing data and a static stopping procedure that generalized from decoding curves estimated from training data.

Furthermore, the study explored whether an individualized approach that selected the optimal stimulus condition for each participant could outperform the globally best-performing condition.

Another central objective was cross-decoding, that is, evaluating whether a decoder trained on one stimulus condition could accurately classify c-VEP responses elicited by a different condition. This was made possible by the reconvolution CCA method [55], which, in theory, supports generalization across any binary stimulus sequence [32]. Additionally, the analysis explored whether a model trained on EEG data from the VEP task could generalize to the c-VEP task.

Specifically, given training data $\mathbf{X}$, labels $\mathbf{y}$, and corresponding structure matrices $\mathbf{M}$, equation (1) was used to derive a class-invariant spatial filter $\mathbf{w}$ and a temporal response function $\mathbf{r}$. For testing, structure matrices $\mathbf{M}$ were computed based on the test stimulation sequences, and equation (2) was applied to predict responses from the test data $\mathbf{X}$. Importantly, the temporal response function $\mathbf{r}$ is estimated separately for rising and falling edges within binary sequences, enabling prediction of template responses for any, including previously unseen, binary stimulus sequences.

Finally, participant-reported eyestrain was analyzed to assess the subjective comfort of different stimulus conditions.

For each of these analysis, a two-way repeated-measures ANOVA (RM-ANOVA) was used to test for statistically significant effects of two factors on ITR: Code family (mseq, debruijn, golay, gold, gold-set) and modulation (original, modulated). Pair-wise post-hoc analyses were carried out using a one-sided paired t-test.

2.6. Data and software availability

The experiment was conducted using Python version 3.8.10 and PsychoPy version 2022.1.4 [67]. Additionally, the SART task was presented using PsyToolkit version 3.4.2 [68, 69].

Data analysis was performed using Python version 3.12.8. Data preprocessing was done using MNE version 1.9.0 [70]. Classification was performed using PyntBCI version 1.8.1 [58]. Statistics were done with the statsmodels library version 0.14.4 [71].

All data are available open access via the Radboud Data Repository [72]. The dataset includes all raw EEG and ECG signals, as well as all scripts used for the experiment and analysis. The data are organized using the brain imaging data structure (BIDS) [73] with the extension for EEG data [74], which was done using MNE-BIDS version 0.16.0 [75].

3. Results

3.1. c-VEP BCI inefficiency

The c-VEP BCI performance was analyzed in three ways (figure 1): (1) When using the full 4.2 s trial duration, (2) when maximizing the accuracy on

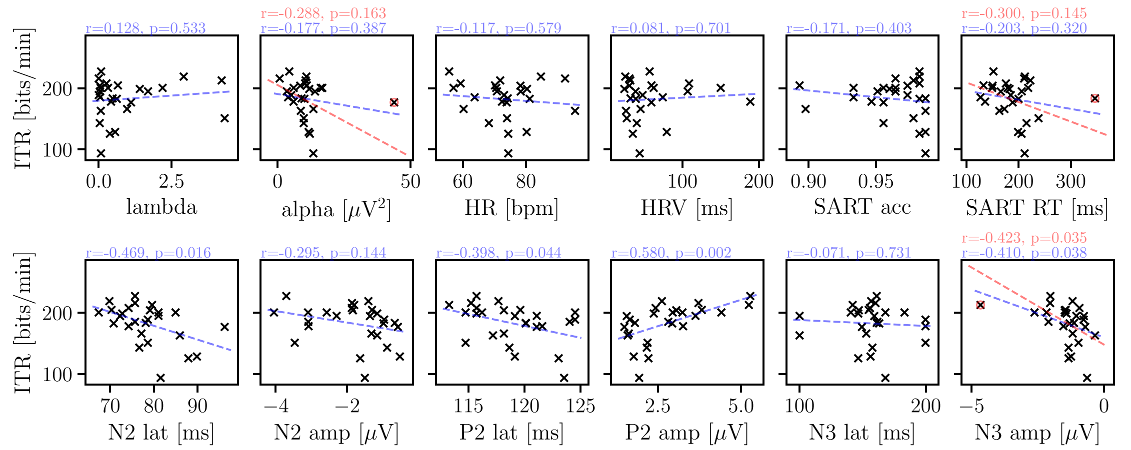

Figure 6. Relationship between predictors and ITR in a c-VEP BCI. The scatter plots show the association between each of the ten predictors and the theoretical information transfer rate (ITR) for individual participants. Predictors include the noise exponent (lambda), the alpha predictor, heart rate (HR), heart rate variability (HRV), accuracy and response time (RT) in the sustained attention to response task (SART), and the latency (lat) and amplitude (amp) of the N2, P2, and N3 components of the flash visual evoked potential (VEP). Blue dashed lines represent ordinary least-squares linear regression fits including all participants, while red dashed lines indicate regression fits after outlier removal. Pearson's correlation coefficient (r) and associated p-value are reported in blue including all participants and in red after outlier exclusion. Red circles mark data points (i.e., participants) identified as outlier.

estimated decoding curves as a theoretical upper-bound, and (3) when maximizing the accuracy using a static stopping procedure. The BCI performance was defined as classification accuracy, trial duration, and ITR. Note, this analysis was limited to the c-VEP data recorded with the typically used circularly shifted m-sequence.

In the full trial analysis, on average, all participants reached a perfect accuracy, except two individuals (sub-01 and sub-09) who reached 0.969. As this analysis used the full 4.2 s available data, no variance is observed in the trial duration. Together, this leads to ITR values ranging between 56.5 and 60.0 bits/min.

In the theoretical analysis, all participants reached a perfect accuracy, except for one individual (sub-01) who reached 0.969. Importantly, contrary to the full trial analysis, the trial duration now varied between 500 ms and 2.5 s, leading to ITR values ranging between 93.5 and 227.3 bits/min.

Finally, in the static stopping analysis, only ten individuals reached perfect classification accuracy, overall ranging between 0.906 and 1.000. The obtained trial duration ranged between 900 ms and 3.4 s. Together, the estimated ITR ranged between 67.2 and 182.3 bits/min.

These results demonstrate that all participants in this study achieved near-perfect classification accuracy. However, while some required minimal data to do so, others needed substantially more. This variability contributed to differences in the obtained ITR across participants, highlighting individual differences in the efficiency of the c-VEP BCI.

3.2. c-VEP BCI performance prediction

Acknowledging the inherent variance in the efficiency of a c-VEP BCI, this study aimed to identify factors that predict performance variability across participants. Ten potential predictors were analyzed, including the noise floor exponent, alpha amplitude, HR and HRV, all estimated from a pre-session eyes-open resting-state task. Additionally, the accuracy and RT were assessed in a post-session sustained attention task. Finally, six predictors were extracted that described the latency and amplitude of the N2, P2, and N3 components of the flash-VEP, measured during a pre-session VEP task.

The relationship between each predictor and the theoretical ITR (figure 1) was assessed using Pearson's product-moment correlation coefficient. In addition, ordinary least-squares (OLS) linear regression was performed for each predictor to estimate its contribution to ITR variability. These analyses included all 26 participants as data points (figure 6, blue). Data points exceeding the 99.9% distribution threshold (i.e., z-score $>$ 3.29) were classified as outliers and removed. If a predictor contained outliers, the correlation and regression analyses were repeated (figure 6, red).

Of the ten predictors, only four showed significant correlations with c-VEP BCI performance ($p < .05$). First, shorter N2 latency was associated with higher ITR ($r = -0.469, p = .016$). Second, shorter P2 latency led to higher ITR ($r = -0.398, p = .044$). Third, larger P2 amplitude correlated with higher ITR ($r = 0.580$, $p = .002$). And fourth, smaller N3 amplitude was linked to higher ITR ($r = -0.410$, $p = .038$), a relationship that slightly strengthened after the removal of one outlier ($r = -0.423, p = .035$).

These results suggest that inter-individual differences in flash-VEP characteristics may contribute to variations in c-VEP BCI performance. Specifically, these factors could help explain why the c-VEP BCI

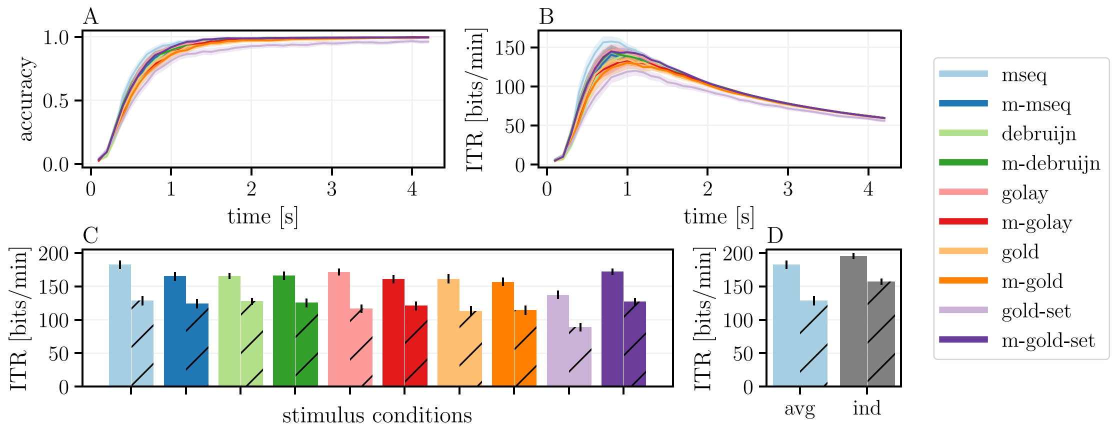

Figure 7. The performance of c-VEP BCI across stimulus conditions. Depicted are the grand average 4-fold cross-validated decoding results across stimulus conditions. (A, B) The decoding curves are shown with accuracy (A) and information transfer rate (ITR, B). (C) The ITR across stimulus conditions obtained theoretically (no stripes), or using static stopping (stripes). (D) The ITR for the best universal stimulus condition (m-sequence) and as optimized per individual, both obtained theoretically (no stripes), or using static stopping (stripes).

using circularly shifted m-sequences performed exceptionally well for certain individuals, and less so for others.

3.3. c-VEP BCI stimulus optimization

Given the observed performance variations across participants and the inter-individual variability in flash-VEP parameters, the full set of ten stimulus conditions was analyzed. The c-VEP dataset contained five distinct code families, each in both its original and modulated forms, yielding ten conditions.

First, theoretical decoding performance and static stopping performance were assessed (figure 7 A,B,C). The grand average results for each stimulus condition are summarized in table 1.

For the theoretical analysis, the highest ITR was achieved by the m-sequence with 182.4±6.2 bits/min, followed by m-gold-set (172.1±4.8), golay (171.7±5.2), m-debruijn (165.8±6.2), debruijn (165.6±4.3), m-mseq (165.0±6.6), gold (161.2±6.9), m-golay (160.6 ±6.2), m-gold (156.7±6.3), and gold-set (137.2±6.0). A similar ranking was observed for static stopping performance, where the m-sequence again led with 128.3±7.0 bits/min, followed by debruijn (127.4±5.2), m-gold-set (126.9±5.6), m-debruijn (125.4±6.6), m-mseq (124.0±6.8), m-golay (120.8±6.5), golay (116.6±6.2), m-gold (114.3±6.9), gold (113.0±7.5), and gold-set (89.0±6.2).

An RM-ANOVA assessed the influence of code family (mseq, debruijn, golay, gold, gold-set) and modulation (original, modulated) on static stopping ITR. Results showed a significant main effect of code family ($F(4) = 7.33$, $p < .001$) but no significant effect of modulation ($F(1) = 3.28$, $p = .082$). However, a significant interaction was found between code family and modulation ($F(4) = 7.24$, $p < .001$).

Post-hoc analysis revealed a significant effect among original code families ($F(4) = 11.84$, $p < .001$), but no significant effect among modulated ones ($F(4) = 1.48$, $p = .214$). Among the original codes, the m-sequence did not significantly outperform the second-best code ($p = .441$), while the gold-set yielded the significantly lowest performance ($p = .002$).

Second, it was investigated whether selecting an individually optimal stimulus condition would yield better performance than using the globally optimal stimulus condition (figure 7 D). The m-sequence was the best-performing code overall, with an ITR of 182.5 and 128.3 bits/min for theoretical and static stopping analyses, respectively. When selecting the individually optimal codes, performance significantly improved to 195.6 and 157.1 bits/min ($p < .001$).

For theoretical ITR, mseq was optimal for 12 participants, followed by m-mseq (3), golay (3), m-gold-set (2), m-golay (2), gold (2), m-debruijn (1), and m-gold (1). For static stopping ITR, mseq was best for 7 participants, followed by m-gold-set (4), gold (4), m-gold (3), m-mseq (2), m-debruijn (2), m-golay (2), debruijn (1), and golay (1).

Third, to assess the ease of decoding, the auto-correlation function was analyzed of both the binary stimulus sequences and the real-valued template responses as used in the template-matching classifier (figure 8). Three key observations emerged: (1) the correlation properties of the stimulus codes do not directly translate to the response domain, (2) code modulation has only a minor effect on correlation, and (3) while correlation varies substantially across code families in the stimulus domain, differences are less pronounced in the response domain.

Fourth, the comfort levels of the different stimulus conditions were evaluated (figure 9). An RM-ANOVA revealed a significant main effect of modulation ($F(1) = 33.23$, $p < .001$), while no significant effect of code family was observed ($F(4) = 0.41$, $p = .805$). However, a significant interaction between code family and modulation was found ($F(4) = 5.30$, $p < .001$).

Table 1. The performance of c-VEP BCI across stimulus conditions. Listed are the grand average 4-fold cross-validated decoding results across stimulus conditions, and their standard deviations. Listed are the classification accuracy (Acc), trial duration in seconds (Dur) and information transfer rate (ITR) in bits/min. Results are shown for both the theoretical (T) as well as static stopping (S) analysis. Note, the trial duration (Dur) does not incorporate the inter-trial time, while the ITR does.

		mseq	m-mseq	debruijn	m-debruijn	golay	m-golay	gold	m-gold	gold-set	m-gold-set
Acc	**T**	99.9±0.1	99.8±0.2	99.8±0.2	99.9±0.1	99.9±0.1	99.8±0.2	99.4±0.4	99.6±0.3	97.7±0.9	99.6±0.2
	S	97.4±0.5	97.5±0.5	97.8±0.5	97.6±0.5	97.0±0.5	97.8±0.4	96.5±0.8	97.0±0.8	93.6±1.2	96.8±0.6
Dur	**T**	0.96±0.1	1.15±0.1	1.08±0.1	1.13±0.1	1.07±0.1	1.19±0.1	1.22±0.1	1.24±0.1	1.52±0.1	1.01±0.1
	S	1.63±0.1	1.70±0.1	1.59±0.1	1.68±0.1	1.84±0.2	1.78±0.1	1.99±0.2	1.95±0.2	2.55±0.2	1.56±0.1
ITR	**T**	182.5±6.2	165.0±6.6	165.6±4.3	165.8±6.2	171.7±5.2	160.6±6.2	161.7±6.9	156.7±6.3	137.8±6.0	172.1±4.8
	S	128.3±7.0	124.0±6.8	127.4±5.2	125.4±6.6	116.6±6.2	120.8±6.5	113.0±7.5	114.3±6.9	89.0±6.2	126.9±5.6

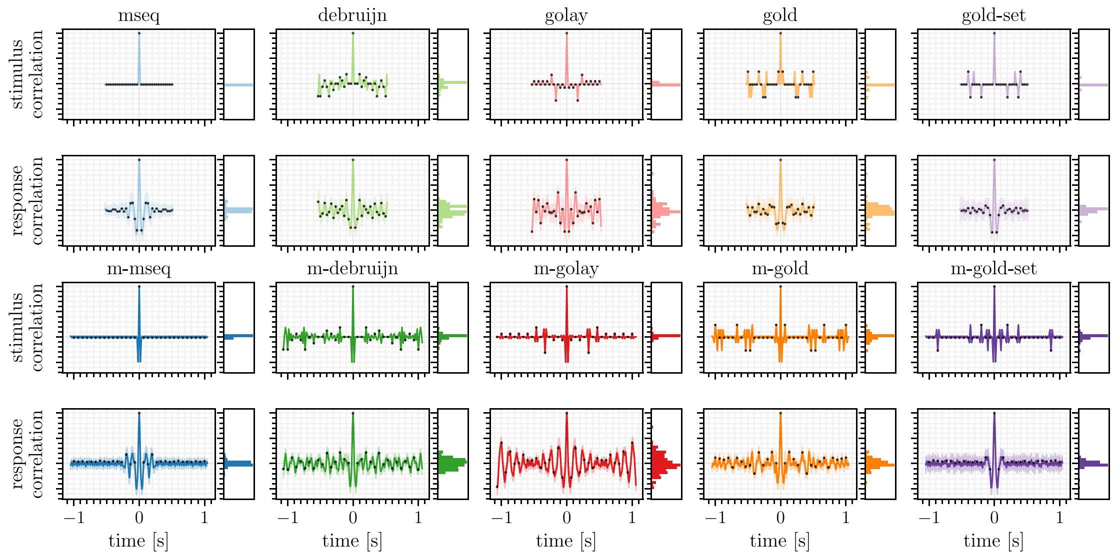

Figure 8. The auto-correlation of a bit-sequence and its c-VEP response across stimulus conditions. (A,C) Depicted is the auto-correlation of the binary stimulus sequences for the original (A) and modulated (C) sequences. (B,D) Grand average and standard deviation of the auto-correlation of the template responses corresponding to the sequences in (A) and (C), for the original (B) and modulated (D) stimuli, respectively. Black dots indicate the circularly shifted codes used in the experiment. For the gold-set and m-gold-set, this figure displays the auto-correlation of a single code, whereas the experiment presented a set of unique codes. Note that the modulated sequences are twice as long, but during the experiment, multiple repetitions of these cycles were presented to ensure a consistent trial duration across conditions.

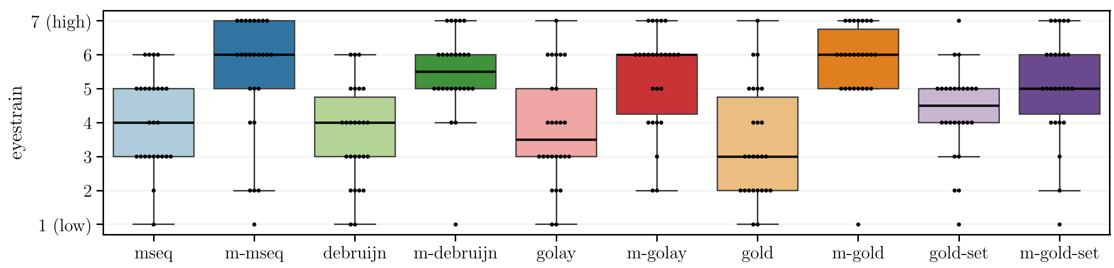

Figure 9. The eyestrain across stimulus conditions as rated by participants. This figure displays participants' subjective judgments of eyestrain, given at the end of the experiment and measured on a 7-point scale. A rating of 7 indicates high eyestrain and low user comfort, while a rating of 1 signifies low eyestrain and high user comfort. The boxes represent the median and inter-quartile range (IQR) of the data distributions, with the whiskers extending to 1.5 times the IQR from the lower and upper quartiles. Black dots indicate individual participants.

Post-hoc analysis revealed a significant effect of modulation across all code families ($p = .001$), except for the gold-set ($p = .032$). Ranked by decreasing eyestrain (i.e., increasing comfort), gold was perceived as having the lowest eyestrain (3.3±1.6), followed by debruijn (3.6±1.4), golay (3.8±1.6), mseq (4.1±1.4), gold-set (4.3±1.3), m-gold-set (5.1±1.5), m-golay (5.3±1.5), m-mseq (5.4±1.8), m-debruijn (5.5±1.3), and m-gold (5.8±1.2).

Finally, to assess the generalizability of responses across stimulus conditions, a cross-decoding analysis was conducted, that is, a transfer learning across stimulus conditions (figure 10). First, the VEP data was used as donor data to calibrate a model, which was then applied to each stimulus condition as a receiver. Second, each c-VEP stimulus condition served as both donor and receiver in all possible pairwise combinations.

Several key observations emerged from this cross-decoding analysis. First, training on VEP data resulted in poor accuracy, not exceeding 0.32. Notably, accuracy was higher for original sequences compared to modulated ones. Second, in general, stimulus conditions performed weakly as donors, achieving accuracy between 0.53 and 0.85, with the exception of the gold-set, which proved to be a strong donor with an average accuracy of 0.97 across receiving conditions. Third, generalization between modulated sequences was highly effective, yielding accuracies between 0.97 and 1.00, whereas generalization between original sequences varied more widely between 0.25 to 0.99. Similarly, generalization from an original to a modulated sequence and vice versa varied widely between 0.26 to 0.99. Finally, no single condition emerged as a consistently strong receiver, with accuracy ranging between 0.64 and 0.85 across receiving conditions.

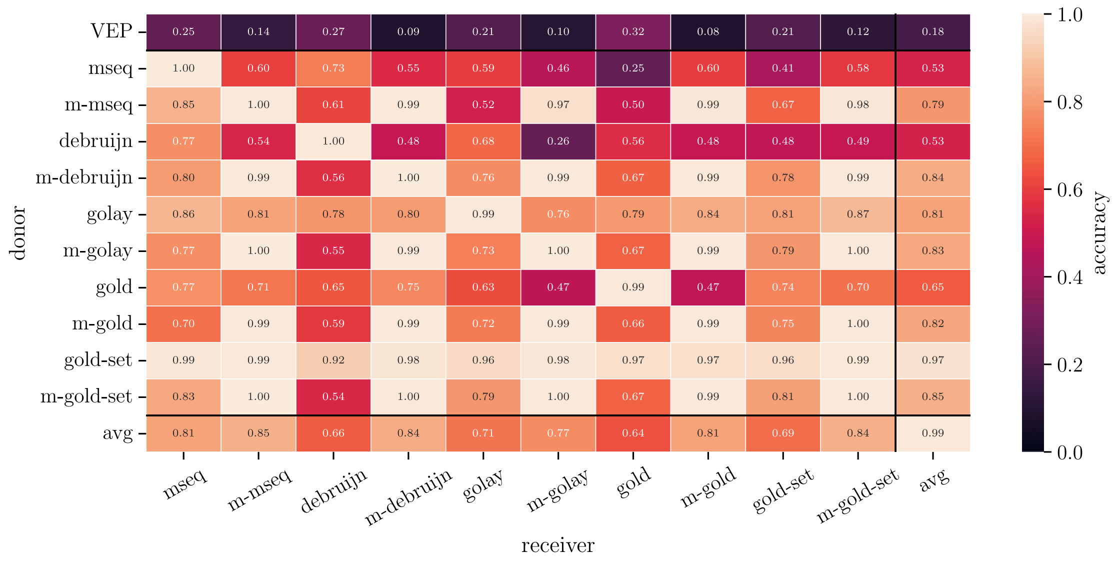

Figure 10. The classification accuracy of cross-decoding stimulus conditions. The figure presents the 4-fold cross-validated cross-condition decoding results, where a model is trained on one stimulus condition (donor) and tested on another (receiver). The last column and last row display the average accuracy for donors and receivers, respectively. The diagonal entries represent standard within-condition decoding, with its average shown in the final row and column. The top row illustrates cross-decoding results when training on data from the VEP task and decoding each of the c-VEP stimulus conditions. In this analysis, all available training and testing data were used.

4. Discussion

This study had three primary objectives. First, it aimed to demonstrate that some degree of BCI inefficiency exists in c-VEP BCIs by highlighting significant variability in efficiency across participants. Second, it aimed to identify factors contributing to this variability, offering insights into the factors underlying individual differences. And third, it aimed to explore the c-VEP stimulus protocol and to tailor it to individual users, by evaluating a diverse set of commonly used binary stimulus sequences.

4.1. c-VEP BCI inefficiency

This study assessed the presence of BCI inefficiency in c-VEP BCI. If one sticks to the traditional definition of BCI inefficiency, that a subset of users cannot achieve reliable control, then the statement does not apply to c-VEP BCI. In this study, 100% of users reached perfect or near-perfect classification accuracy, consistent with previously reported findings [16]. Therefore, the commonly cited claim about BCI inefficiency should be rephrased as "BCI control fails for 15–30% of users *of MI-based BCI*".

However, if the definition of BCI inefficiency is considered at a continuous spectrum, where users achieve varying degrees of control, then inefficiency also exists in c-VEP BCI. That is, while all users could maintain near-perfect accuracy, their obtained ITRs varied substantially due to differences in speed. This observation challenges the claim that BCI inefficiency is absent in c-VEP BCIs [16].

4.2. c-VEP BCI performance prediction

This study identified four neurophysiological predictors of c-VEP BCI performance: N2 latency, P2 latency and amplitude, and N3 amplitude. These components, which are dominant features of the flash-VEP [66], were found to significantly correlate with BCI performance. Consistent with these findings, previous studies have reported similar inter-individual variations in amplitude and latency [21, 24]. Additionally, factors such as anatomy [23], age [22, 66], and sex [22, 24] have been shown to influence the flash-VEP. Notably, a c-VEP BCI study observed a trend suggesting that females outperformed males, though the results were not statistically significant [76].

Beyond these three significant predictors, a strong negative correlation, although insignificant, was found between c-VEP BCI performance and alpha peak activity recorded during a pre-session, eye-open resting-state task. This indicates that lower alpha activity is associated with better c-VEP BCI performance. This aligns with previous research showing that alpha activity is linked to inhibitory responses and information suppression [62].

Interestingly, the well-established sensorimotor mu predictor in MI-based BCIs showed a positive correlation with BCI performance [11]. This distinction is logical, because lower alpha activity supports effective processing and an alert state, but a strong mu rhythm allows for greater desynchronization during imagined movement, enhancing the SNR in MI-based BCIs.

A similar contrast between high and low alpha activity could potentially be beneficial under other code-modulated paradigms. For instance, gaze-independent c-VEP BCIs [77] and BCIs relying on the code-modulated auditory evoked potential (c-AEP) [78] may rely on lateralization of alpha power induced by the spatial covert attention task. Specifically, alpha desynchronization (i.e., excitation) is expected at the contra-lateral task-positive site, whereas alpha synchronization (i.e., inhibition) is expected at the ipsilateral task-negative site.

4.3. c-VEP BCI stimulus optimization

This study conducted a comprehensive evaluation of a large set of binary stimulus sequences for c-VEP BCI, seeking to determine which factors contribute to optimal performance and user comfort. A key finding was that the code family had a significant effect on ITR, while modulation did not. Among all tested sequences, the circularly shifted m-sequence consistently outperformed the others, reinforcing its suitability for c-VEP BCI applications.

These findings are consistent with previous research demonstrating the superior performance of m-sequences compared to other stimulus sequences [25, 43], and reflect the dominant use of the m-sequence for c-VEP BCI [6]. However, they contrast with studies suggesting that de Bruijn and Golay sequences outperform m-sequences [34, 35]. These discrepancies may highlight the inter-individual differences within the tested participant population, emphasizing the need for personalized approaches in c-VEP BCI optimization.

Beyond classification performance, this study also examined the subjective comfort associated with different stimulus conditions. While code family did not significantly affect reported eyestrain, modulation did, where participants consistently rated modulated sequences as less comfortable than their original counterparts. This finding underscores the importance of considering not only performance but also user experience when designing BCI paradigms.

Furthermore, a critical question in BCI research is whether a universal stimulus exists that provides the best performance for all users, or whether tailoring the stimulus to individual characteristics could further optimize outcomes. This study showed that individualized stimulus selection led to a significant increase in ITR, surpassing the performance of the overall best sequence, here the circularly shifted m-sequence. This aligns with prior research that demonstrated performance improvements when selecting among a smaller subset of sequences, including m-sequences, Gold codes, and Barker codes [33].

Finally, this study investigated the feasibility of cross-decoding, an essential capability for optimizing c-VEP BCI stimulation without requiring additional extensive calibration data for each new participant. While generalization was successful between modulated sequences, other cross-decoding attempts revealed substantial variability in classification accuracy.

This variability in cross-decoding may stem from differences in event distributions across stimulus conditions, and with that a difference in the distribution of inter-stimulus intervals (ISIs). Notably, modulated codes are characterized by restricted 1-bit and 2-bit run lengths and thus a more uniform distribution of ISIs. Instead, original codes include a broader distribution ranging from 1-bit to 6-bit run lengths. This has two consequences: (1) There are more events per unit time in modulated sequences, and (2) there might be more non-linearity introduced in the original codes due to more variable events, which are not yet accounted for in the overall linear reconvolution response modeling.

In line with these results, previous studies have demonstrated that within a code family (i.e., those sharing the run length distribution), generalization to unseen sequences is possible, for instance, one Gold code set could reliably predict responses to another unseen Gold code set [55]. More recently, approaches leveraging temporal response functions (TRFs), akin to the reconvolution framework, have been shown to facilitate robust prediction of responses to novel unseen white-noise stimuli [8]. These findings highlight the potential for cross-decoding strategies, paving the way for more flexible and adaptive c-VEP stimulus implementations.

A particularly intriguing observation of this study is the possibility that different stimulus sequences may be optimal for calibration (i.e., donor) versus decoding (i.e., receiver). During the calibration phase, the primary objective is to maximize the SNR of the attended sequence, ensuring a clean and well-defined parameter estimation of the flash-VEP. However, during real-time decoding, the goal shifts beyond maximizing SNR to also enhancing contrast between competing stimulus sequences, thereby facilitating classification.

This rationale may align with previous research showing that longer ISIs can improve SNR, as demonstrated with burst-codes [48]. If different sequences are indeed better suited for these distinct phases, this also has major implications for instantaneous and zero-training approaches that perform calibration and decoding on the same current trial [32, 60]. Future research should further explore how the stimulus protocol contributes to easier calibration as well as better decoding, besides the decoding algorithm.

4.4. Limitations and future directions

The findings of this study should be interpreted with caution. Firstly, this study used a limited sample size of 26 participants. The reported results must be replicated and validated on larger and independent datasets. This underscores the importance of open-access data sharing, which facilitates replication efforts and accelerate progress in the field, such as MOABB [79]. Additionally, this study highlights the importance of including an eyes-open resting-state block prior to experiments, as it may serve as a valuable baseline for understanding individual differences.

Secondly, this study used a theoretical upper bound by estimating a decoding curve from the testing data, while also employing a static stopping procedure that maximized accuracy using a decoding curve from training data. Notably, the static stopping procedure did not reach the theoretical upper bound, suggesting that the practical performance could be further optimized. Recent studies have introduced dynamic stopping strategies for evoked responses [80, 81]. Future research should explore how these adaptive stopping techniques might enhance efficiency and reliability in c-VEP BCI applications.

Thirdly, this study is the first to empirically compare a large set of stimulus conditions. However, the analysis was limited to binary stimuli. The field is more and more moving towards non-binary [26] and white-noise stimuli [7]. Such stimuli have shown improved performance and user comfort. Similarly, there is a move towards stimuli with lower contrast levels and localized contrast patches [82, 83].

Fourthly, another key consideration in this study was the comparison between original and modulated codes. While modulation effectively constrained the run length distribution, it also doubled the stimulus cycle length. While all stimulus conditions were presented in trials of equal duration, as a consequence the original codes contained twice as many cycles as modulated codes. This discrepancy may confound comparisons of classification performance and speed.

Fifthly, this study focused on healthy users and revealed notable inter-individual differences within this population. A patient population, however, may exhibit distinct response characteristics [84]. Additionally, the choice of recording modality, such as EEG versus stereoEEG, can significantly influence the SNR and consequently the performance of c-VEP-based BCIs [85]. These domain-specific variations could benefit greatly from an online adaptive stimulus and decoding protocol.

Finally, while much of the BCI field has focused on advancing sophisticated decoding methods to optimize performance, the underlying neural mechanisms of c-VEP remain poorly understood. Many decoding models assume a linear relationship between the stimulus and response. That is, a typical assumption is that each flash elicits an identical EEG response regardless of when and where it was presented. However, this assumption overlooks neural adaptation, where the response to consecutive flashes is dampened proportional to the ISI. A better understanding of the (non-linear) neural mechanisms underlying c-VEP is essential to improving both decoding strategy and stimulus paradigm.

5. Conclusion

This study investigated the presence, several predictors, and potential solutions for BCI inefficiency in c-VEP BCIs. While all participants reached near-perfect classification accuracy, this study demonstrated a significant inter-individual variability in classification speed and ITR, highlighting the existence of c-VEP BCI inefficiency. Additionally, this study identified four neurophysiological factors that correlated with the observed variability in c-VEP BCI performance: the N2 latency, the P2 latency and amplitude, and the N3 amplitude. Finally, the study compared a large body of binary stimulus sequences and found that the circularly shifted m-sequence remains the best binary stimulus overall. However, performance could be further improved when tailoring the stimulus protocols to individual users.

This study supports the idea that a one-size-fits-all approach may not be optimal. Individuals display a wide variety of characteristics that should be considered when designing efficient BCI systems. These findings call for further research on individualized and adaptive BCI stimulus protocols and decoding paradigms.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.34973/6dw9-0924.

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